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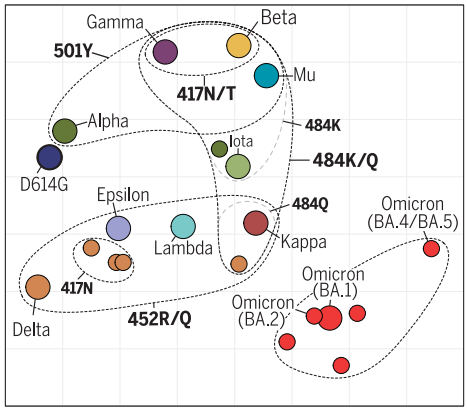
Mapping SARS-CoV-2 antigenic relationships and serological responses

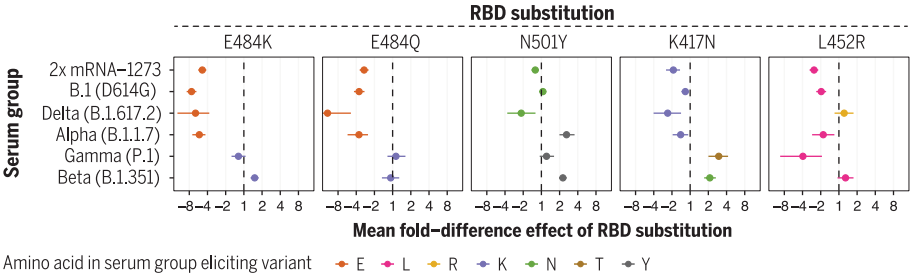
Samuel H. Wilks*†, Barbara Mühlemann†, Xiaoying Shen†, Sina Türeli, Eric B. LeGresley, Antonia Netzl, Miguela A. Caniza, Jesus N. Chacaltana-Huarcaya, Victor M. Corman, Xiaoju Daniell, Michael B. Datto, Fatimah S. Dawood, Thomas N. Denny, Christian Drosten, Ron A. M. Fouchier, Patricia J. Garcia, Peter J. Halfmann, Agatha Jassem, Lara M. Jeworowski, Terry C. Jones, Yoshihiro Kawaoka, Florian Krammer, Charlene McDanal, Rolando Pajon, Viviana Simon, Melissa S. Stockwell, Haili Tang, Harm van Bakel, Vic Veguilla, Richard Webby, David C. Montefiori*, Derek J. Smith*

INTRODUCTION: Vaccination has greatly reduced the disease burden of SARS-CoV-2. However, since late 2020, variants have emerged that are able to escape immunity from vaccination and previous infections, including B.1.351 (Beta), P.1 (Gamma), B.1.617.2 (Delta), and B.1.1.529 (Omicron and its descendants). In combination with vaccination, infections with different variants form the basis of current population immunity against SARS-CoV-2.

RATIONALE: Understanding the antigenic relationships among SARS-CoV-2 variants, the substitutions that cause antigenic change, and how the immune response is shaped by previous infections, is crucial for understanding the evolution of the virus, determining whether new variants avoid neutralization from antibodies induced by current vaccines, and evaluating the need for vaccine updates.

To this end, we analyzed patterns of neutralization and cross-reactivity among a panel of 21 SARS-CoV-2 variants and 15 groups of human sera obtained from individuals after primary infection with one of 10 different variants or after D614G or B.1.351 vaccination. First, we sought to understand patterns of cross-reactivity and response breadth in postvaccination responses 4 weeks and >3 months after second or third vaccine doses. Then, we used antigenic cartography to visualize antigenic relationships between 21 SARS-CoV-2 variants and experimentally test point mutations to investigate the drivers of the antigenic changes observed in the antigenic map. Lastly, we investigated how serological reactivity post-infection relates to the primary-exposure variant.





Serological responses following exposure to different SARS-CoV-2 variants. (Top) Antigenic map showing antigenic relatedness of SARS-CoV-2 variants. Variants are grouped by amino acid substitutions present at positions 417, 452, 484, and 501, with an additional grouping for the six Omicron variants. (Bottom) Average effect size of single substitutions at different RBD positions on sera from individuals with different primary exposures, colored by the amino acid present at the respective RBD position in the eliciting variant.

RESULTS: Quantifying changes in cross-reactivity and response breadth after vaccination, our results show the largest increase between 4 weeks and >3 months after a second dose. In particular, we found that the main short-term effect of the third vaccination was to boost the magnitude of a response that had already become more cross-reactive rather than to generate significant additional breadth of cross-reactivity.

Using antigenic cartography, we inferred and subsequently experimentally tested our inference that antigenic differences among pre-Omicron variants are primarily caused by substitutions at spike-protein positions 417, 452, 484, and 501 (see figure, top). The experimental effect of these substitutions was largely consistent with those inferred from the map.

We also found that sensitivity to these substitutions varied greatly between individuals infected with different variants. These differences are consistent with substantial changes in immunodominance of different spike regions, depending on the variant an individual was first exposed to and the amino acid present at these positions in the eliciting variant (see figure, bottom). For example, whereas sera of individuals exposed to D614G, B.1.1.7, and B.1.617.2 variants are sensitive to changes at position 484, sera of individuals exposed to B.1.351 and P.1 are not.

CONCLUSION: Our results provide a comprehensive analysis of the antigenic variation between SARS-CoV-2 variants and the development of the immune response after infection or vaccination. The large antigenic effect of a small number of substitutions in the receptor-binding domain (RBD) of SARS-CoV-2 is similar to the pattern observed for seasonal influenza viruses, for which major antigenic changes are often associated with single or double substitutions. These substitutions in the SARS-CoV-2 RBD not only allow the virus to escape from preexisting immunity but also influence the regions of the spike protein that the immune response targets. Depending on their infection history, different individuals can thus be sensitive to substitutions in different regions of the spike protein. As individuals increasingly experience multiple infections, choosing vaccine immunogens on the basis of immunodominance considerations may be an important aspect in ensuring high vaccine efficacy across populations with different patterns of preexisting immunity. ■

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Cite this article as S. H. Wilks et al., Science 382, eadj0070 (2023). DOI: 10.1126/science.adj0070

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Mapping SARS-CoV-2 antigenic relationships and serological responses

Samuel H. Wilks[1]*†, Barbara Mühlemann[2,3]†, Xiaoying Shen[4,5]†, Sina Türeli[1], Eric B. LeGresley[1], Antonia Netzl[1], Miguela A. Caniza[6], Jesus N. Chacaltana-Huarcaya[7], Victor M. Corman[2,3], Xiaoju Daniell[4], Michael B. Datto[8], Fatimah S. Dawood[9], Thomas N. Denny[5], Christian Drosten[2,3], Ron A. M. Fouchier[10], Patricia J. Garcia[11], Peter J. Halfmann[12], Agatha Jassem[13], Lara M. Jeworowski[2], Terry C. Jones[1,2,3], Yoshihiro Kawaoka[12,14,15,16], Florian Krammer[17,18], Charlene McDanal[4], Rolando Pajon[19], Viviana Simon[17,18,20,21], Melissa S. Stockwell[22], Haili Tang[4], Harm van Bakel[23], Vic Veguilla[9], Richard Webby[24], David C. Montefiori[4,5]*, Derek J. Smith[1]*

During the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic, multiple variants escaping preexisting immunity emerged, causing reinfections of previously exposed individuals. Here, we used antigenic cartography to analyze patterns of cross-reactivity among 21 variants and 15 groups of human sera obtained after primary infection with 10 different variants or after messenger RNA (mRNA)–1273 or mRNA-1273.351 vaccination. We found antigenic differences among pre-Omicron variants caused by substitutions at spike-protein positions 417, 452, 484, and 501. Quantifying changes in response breadth over time and with additional vaccine doses, our results show the largest increase between 4 weeks and >3 months after a second dose. We found changes in immunodominance of different spike regions, depending on the variant an individual was first exposed to, with implications for variant risk assessment and vaccine-strain selection.

Since the beginning of the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic, the virus has caused more than 766 million cases and 6.9 million deaths (*1*). During the first year of the pandemic, circulation was dominated by the B.1 variant, characterized by the D614G substitution in the spike protein, which imparted increased infectivity and transmissibility in vitro and in animal models (*2*, *3*) but did not escape serum neutralization (*4*). Since then, multiple variants have circulated widely, with five of them—B.1.1.7 (Alpha), B.1.351 (Beta), P.1 (Gamma), B.1.617.2 (Delta), and B.1.1.529 (Omicron and descendant sublineages)—categorized as variants of concern by the World Health Organization (WHO) on the basis of evidence of higher transmissibility, increased virulence, and/or reduced effectiveness of vaccines, therapeutics, or diagnostics (*5*). Sublineages arising within the Omicron lineage show considerable antigenic diversity, often being as antigenically different from the early Omicron variant (Omicron BA.1) as BA.1 is from the B.1-like viruses (*6*).

Before the emergence of the Omicron variants, several variants circulating widely in 2021 were antigenically distinct from the B.1-like 2020 viruses, with B.1.351 escaping neutralization by convalescent and postvaccination sera the most (*7*, *8*). Moreover, sera from individuals first infected with B.1.351 and P.1 failed to readily neutralize B.1.617.2, and vice versa, although B.1.617.2 did not show strong escape from convalescent sera after infection with prototype-like variants (*9*, *10*). All variants in the Omicron lineage show substantial escape from postvaccination and convalescent sera after infection with the pre-Omicron variants (*11*, *12*).

Understanding the antigenic relationships among SARS-CoV-2 variants and the substitutions that cause antigenic change is required for evaluating the need for vaccine updates and predicting whether new variants avoid neutralization from antibodies induced by current vaccines. However, with an increasing number of variants, understanding antigenic relationships through neutralization titer data becomes more complex. Antigenic cartography (*13*) is a tool originally developed for the analysis of human seasonal influenza virus antigenic data and has since been used in the analysis of antigenic variation in other pathogens, including avian, equine, and swine influenza viruses (*14*–*18*); flaviviruses (*19*), including dengue viruses (*20*, *21*); lyssaviruses (*22*); and foot-and-mouth disease viruses (*23*). It provides a quantitative and visual summary of antigenic differences among large numbers of variants and is a core component of the biannual influenza virus vaccine-strain selection process convened by the WHO. In this study, we used antigenic cartography to analyze patterns of cross-reactivity among a panel of 21 SARS-CoV-2 variants and 15 groups of human sera obtained from individuals after primary infection with one of 10 different variants or after prototype or B.1.351 primary-series vaccination. We followed this with experimental testing of point mutations to investigate the drivers of the antigenic changes observed and how the effects of subsequent changes on serological reactivity relate to the primary-exposure variant.

Results

Variant reactivity by serum group

We used 207 serum specimens collected from vaccinated or infected individuals (table S1) and titrated in a Food and Drug Administration (FDA)–approved (FDA/CBER Master File 026862) neutralization assay against lentiviral pseudotypes encoding the spike protein of 21 SARS-CoV-2 variants (fig. S1). Titrations included a panel of 15 pre-Omicron variants and 6 Omicron variants. Before the collection of each serum sample, individuals had reported no known previous SARS-CoV-2 infections or vaccination (table S1). The infecting variant was determined by using whole-genome sequencing. The serum specimens are from individuals with infections with D614G (n = 15), B.1.1.7 (Alpha, n = 14), B.1.351 (Beta, n = 19),

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P.1 (Gamma, n = 17), B.1.617.2 (Delta, n = 28), B.1.526+E484K (Iota, n = 6), B.1.637 (n = 3), C.37 (Lambda, n = 4), BA.1 (Omicron, n = 7), and BA.2 (Omicron, n = 1). We also included groups of sera collected from individuals twice vaccinated with a vaccine comprising prototypic SARS-CoV-2 spike (Moderna mRNA-1273) at 4 weeks (n = 32) or >3 months (n = 16) after a second dose, and sera after a third dose of the same vaccine at 4 weeks (n = 26) or >3 months (n = 8). Lastly, we included samples collected from individuals 4 weeks after a second dose of a B.1.351 spike vaccine (Moderna mRNA-1273.351) (n = 11).

The geometric mean titer (GMT) and individual reactivity profiles for 183 sera after the exclusion of 24 outliers with titers indicative of an unreported previous infection are shown in Fig. 1 (see "Excluding outlier sera" section in Materials and methods, and fig. S2). Each serum group exhibited a distinct profile of neutralization against the tested variants and, as expected, homologous serum and virus pairs were among the most potent in each group

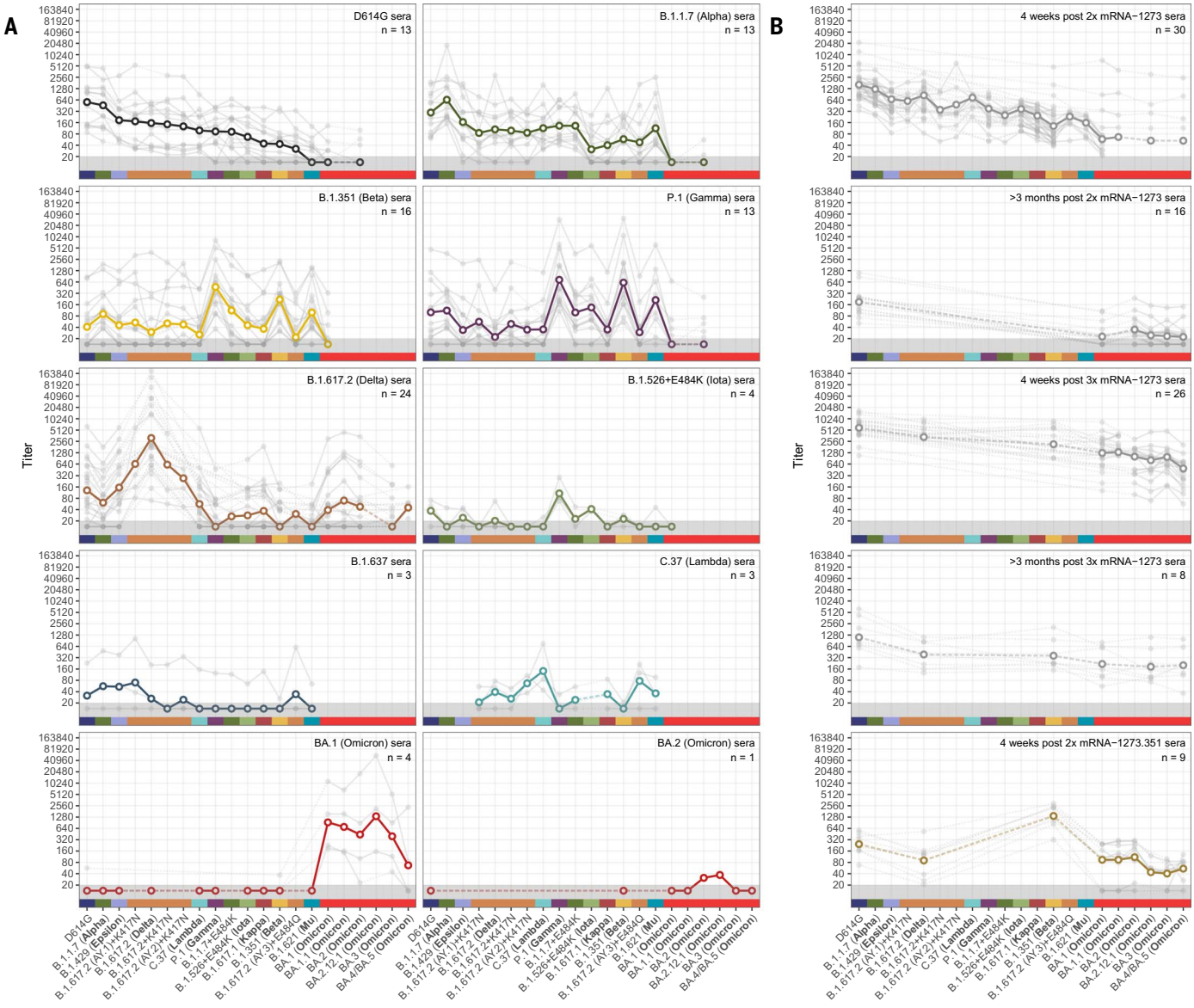

Fig. 1. Neutralization of lentivirus pseudotypes encoding different SARS-CoV-2 spike proteins against different groups of human sera collected after vaccination or primary infection with different variants. Serum groups are split into (**A**) sera elicited by infection with different variants and (**B**) sera elicited by vaccination. Variants are ordered according to geometric mean titer (GMT) in D614G sera [(A), top left], and additional Omicron variants are ordered chronologically. Bold lines with empty circles show the GMTs calculated after estimated differences in individual response size were removed to mitigate biases in which not all sera from a group were titrated against a particular variant (as described in Materials and methods "Titer analyses" section). Fainter individual lines and solid points show individual serum titers. Points in the gray region at the bottom of the plots show titers and GMTs that fell below the detection threshold of 20. Each panel is labeled according to the respective serum group and color coded as indicated on the x axis. Figure S3 shows titers split by sample source for the five serum groups in which samples came from a mixture of cohorts or agencies. Titer box plots, line plots showing the individual serum titers after accounting for individual effects, and titer fold differences relative to the homologous titer are shown in figs. S4, S5, and S6, respectively. Single-letter abbreviations for the amino acid residues are as follows: A, Ala; C, Cys; D, Asp; E, Glu; F, Phe; G, Gly; H, His; I, Ile; K, Lys; L, Leu; M, Met; N, Asn; P, Pro; Q, Gln; R, Arg; S, Ser; T, Thr; V, Val; W, Trp; and Y, Tyr. In the mutants, other amino acids were substituted at certain locations; for example, D614G indicates that aspartic acid at position 614 was replaced with glycine.

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(although some serum groups included only a small number of sera) (Fig. 1A). The B.1.351 and P.1 serum groups both exhibited similar cross-neutralization of the B.1.351 and P.1 variants, suggesting a shared phenotype consistent with the closely matched receptor-binding domain (RBD) substitutions in these two variants (fig. S1). The Omicron variants showed the greatest escape from sera after second and third mRNA-1273 vaccination. Notably, BA.4/BA.5 titers were less than the other Omicron variants tested in the 4-weeks post–third mRNA-1273 vaccination sera but not in the samples taken directly before a third dose (>3 months after second mRNA-1273) or >3 months after a third dose. Comparable escape was seen for the Omicron and B.1.617.2 variants against sera obtained after two doses of mRNA-1273.351. Most BA.1 sera, and the BA.2 serum sample, showed no detectable reactivity to any pre-Omicron variants. However, for one BA.1 serum sample, in which titers were very high, some low levels of measurable reactivity were measurable for the B.1.351 and D614G variants (Fig. 1A). Overall, our findings relating to relative antigenic escape of the different Omicron variants against different serum groups were consistent with findings in other studies that used both lentivirus pseudotype and live-virus neutralization assays, in which overlap was present (*12*) (fig. S7).

Comparing postvaccination and postinfection sera, less variation among individuals was seen in the mRNA-1273 and mRNA-1273.351 vaccine sera than in the corresponding D614G and B.1.351 convalescent sera, both in terms of response magnitude and the pattern of reactivity seen against the variants (Fig. 1 and fig. S5). This is likely related to the more standardized dose (*24*) and nature of vaccination compared with infection (*25–27*) but might also be due to varying time intervals between infection and specimen collection in the convalescent specimens compared with those in postvaccination specimens (*28*).

Comparing vaccine response breadth

We calculated the breadth of postinfection and postvaccination responses, controlling for differences in titer magnitude by focusing on changes in the pattern of fold drops for each variant relative to the homologous variant (see "Calculating fold-drop differences in vaccine sera" in Materials and methods). We found that titer fold drops relative to D614G in each of the mRNA-1273 vaccination serum groups had a pattern similar to those of D614G convalescent sera, but the size of the fold drops was decreased, corresponding to an increased response breadth (Fig. 2A). Moreover, we found a temporal pattern of increasing response breadth (Fig. 2B), with the largest increase between 4 weeks and >3 months after a second dose and, to a lesser extent, between 4 weeks and >3 months after a third dose. In samples taken 4 weeks after a third vaccine dose, although titers were strongly boosted, breadth remained very similar to that measured in >3 months after two-dose mRNA-1273 samples taken directly before the third dose. For the B.1.351 postinfection and postvaccination groups, although titers were higher in the mRNA-1273.351 vaccination group compared with those of B.1.351 convalescent sera, we did not find evidence for a significant difference in the breadth of cross-reactivity (fig. S8).

Antigenic cartography

To visualize and quantify how the different variants relate to each other antigenically, we used the titrations shown in Fig. 1 to construct an antigenic map, in which antigens and sera are positioned relative to each other such that the distance between them corresponds to the fold drop compared with the maximum serum titer (see "Antigenic cartography" in Materials and methods). To incorporate the information from the different post–mRNA-1273 vaccine serum groups but also account for how their increased cross-reactivity would otherwise underestimate the relative antigenic differences between variants, we scaled distance estimates from these serum groups according to the fold-change difference estimates shown in Fig. 2B. Cross-validation results indicated that the neutralization data could generally be well represented in two dimensions (fig. S9), as shown in Fig. 3A. Overall, the antigenic relationships depicted in this map were robust to assay noise and the exclusion of serum groups and variants (Materials and methods, and figs. S10 to S21). The antigenic distinction between the B.1.617.2 variant and the three B.1.617.2 variants with K417N was, however, found to be predominantly driven by patterns of reactivity in the B.1.617.2 sera (figs. S16 and S18).

The clearest deviation from a two-dimensional (2D) representation of the antigenic relationships was due to the BA.4/BA.5 variant titers, for which antigenic differences fit better in three dimensions (Fig. 3B). Compared with its position in 2D space, the variant occupies a position that is more antigenically distinct from the other Omicron variants and closer to the B.1.617.2 sera. This positioning of BA.4/BA.5 reflects the fold change between BA.1 and BA.4/BA.5 in the B.1.617.2 and BA.1 serum groups (fig. S25). BA.4/BA.5 showed a drop in titers compared with BA.1 in the BA.1 convalescent sera but showed increased titers compared with BA.1 in the B.1.617.2 convalescent sera, possibly because BA.4/BA.5 and B.1.617.2 share the substitutions L452R and T478K. The relative antigenic distances estimated for BA.4/BA.5 were quite robust in relation to pre-Omicron serum groups, but the lower number of Omicron sera did lead to some additional uncertainty in the antigenic distance relative to the BA.1 and BA.2 sera.

Serological reactivity shown by antibody landscapes

Antigenic relationships depicted in the antigenic map provide a summary of how the reactivity of the different serum groups distributes among the variants. Figure 4 extends this visualization to antibody landscapes, where a surface in a third dimension represents an estimate of how the reactivity of each serum group and individual serum varies across antigenic space (see "Construction of the antibody landscapes" in Materials and methods). The x-y plane is depicted by the antigenic map in Fig. 3A, and the height of the landscape over a particular point or variant represents the estimated magnitude of serum reactivity in that antigenic region. Antibody landscapes therefore give an indication of how serum reactivity distributes after exposure to different variants, show how the magnitudes of the responses compare, and predict expected levels of reactivity to variants that have not been titrated (fig. S26).

Consistent with the titers in Fig. 1, the shape of the antibody landscapes is similar for the D614G and B.1.1.7 serum groups, with the highest reactivity centered on the D614G and B.1.1.7 variants. Similarly, the shape of the landscapes generated by the mRNA-1273.351, B.1.351, and P.1 sera is comparable, with highest reactivity centered on the B.1.351 and P.1 variants. The landscape of the B.1.617.2 sera shows a contrasting topology, with the highest reactivity indicated against the B.1.617.2 variant and falling off toward other areas of the map. On average, the B.1.617.2 sera titers against non-B.1.617.2 variants were often lower than predicted from the landscape. This suggests either that the B.1.617.2 sera could discriminate between B.1.617.2 and other variants more than was seen in reverse in the non-B.1.617.2 serum groups or that our measurements of B.1.617.2 serum antibodies against B.1.617.2 were biased toward higher values (fig. S26). Because our data showed a larger difference in reactivity between the B.1.617.2 and D614G variants in B.1.617.2 sera when compared with other sources (fig. S7), we speculate that the latter possibility may be the case. Separately, and in agreement with the titers, the landscapes show that the pre-Omicron sera investigated here would be expected to have markedly reduced reactivity against variants in the Omicron lineage (Fig. 4 and fig. S26). However, we note again that titers against the BA.4/BA.5 variant are less reliably predicted when assuming antigenic relationships in only two dimensions. Landscapes of BA.1 first-infection sera show that the drop-off of titers to largely nondetectable levels against pre-Omicron variants is steeper than that seen in reverse for the pre-Omicron sera landscapes. This could be related to the small number of

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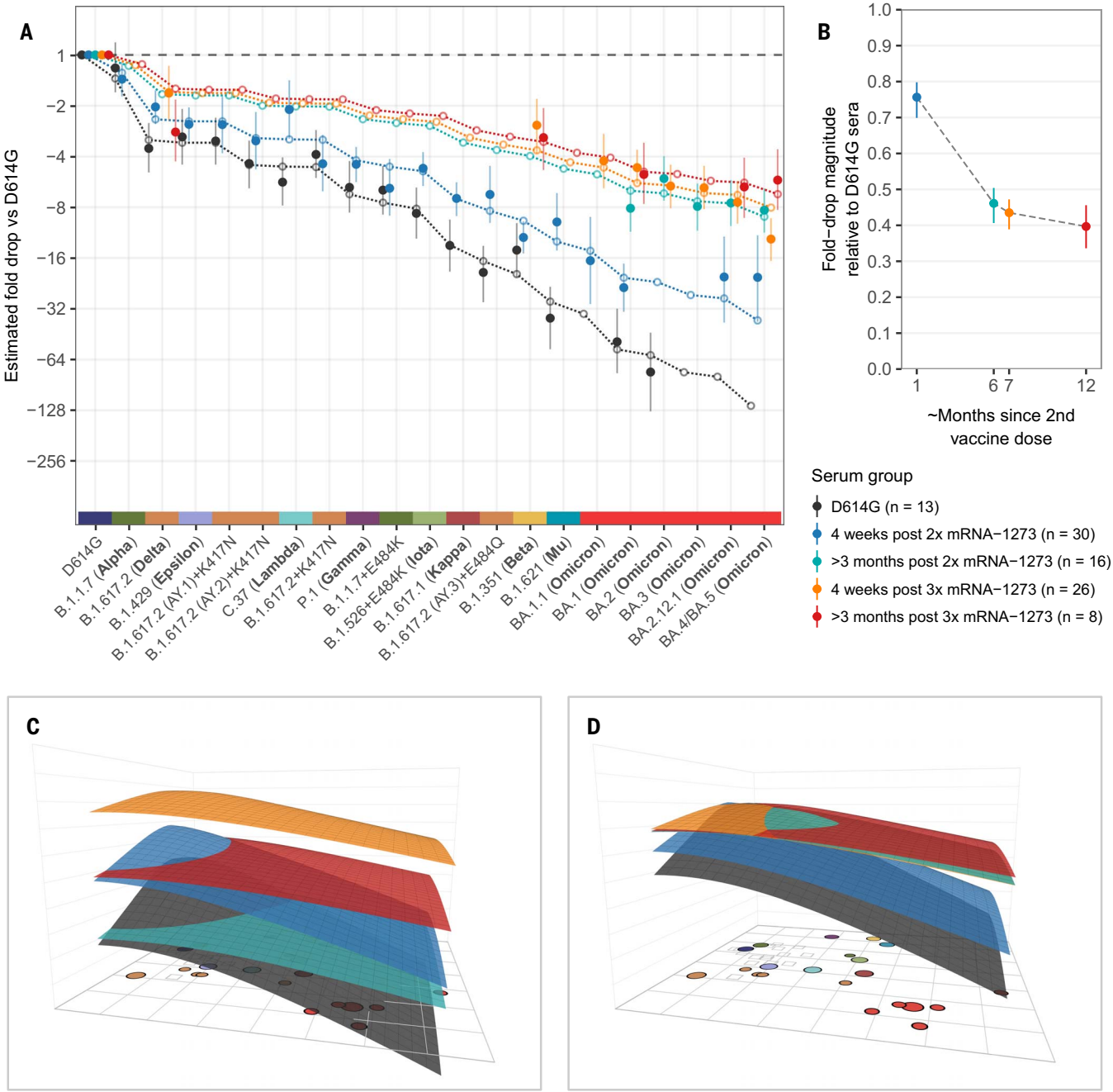

Fig. 2. Comparison of fold drops with different variants in post–D614G infection and post–mRNA-1273 vaccination sera. (**A**) Comparison of different estimates of titer fold-drop responses against different variants. Solid points show the estimate for the mean fold drop compared with the titer for D614G, and solid lines represent the 95% HDI for this estimate. The points for D614G to the left of the plot represent the homologous virus against which fold change for other strains was compared and are therefore fixed at 1. Dotted lines and outline circles show estimates based on a model that assumes a shared overall pattern of fold drops but estimates "slope" differences in the rate of reactivity drop-off seen in the four serum groups (as described in Materials and methods, "Calculating fold-drop differences in vaccine sera"). To aid comparison, points and lines for each of the serum groups are slightly offset from each other in the *x* axis to ensure that points do not directly overlap each other. (**B**) Estimates of fold-drop magnitudes for each mRNA-1273 serum group, relative to the fold-drops seen in the D614G convalescent serum group. Lines show the 95% HDI for each of the estimates, and the position on the *x* axis is proportional to the number of months since a second vaccine dose, assuming an average of 6 months for sera in the >3-months post–2× mRNA-1273 and >3-months post–3× mRNA-1273 groups. (**C**) Antibody landscapes showing how estimates of the mean titer for each of the serum groups in (A) vary across antigenic space. (**D**) Antibody landscapes, as shown in (C), but with a fixed peak titer of 2560 against the D614G variant to highlight differences in titer drop-off slopes. Interactive versions of the landscapes shown in (C) and (D) are available at https://acorg.github.io/mapping_SARS-CoV-2_antigenic_relationships_and_serological_responses.

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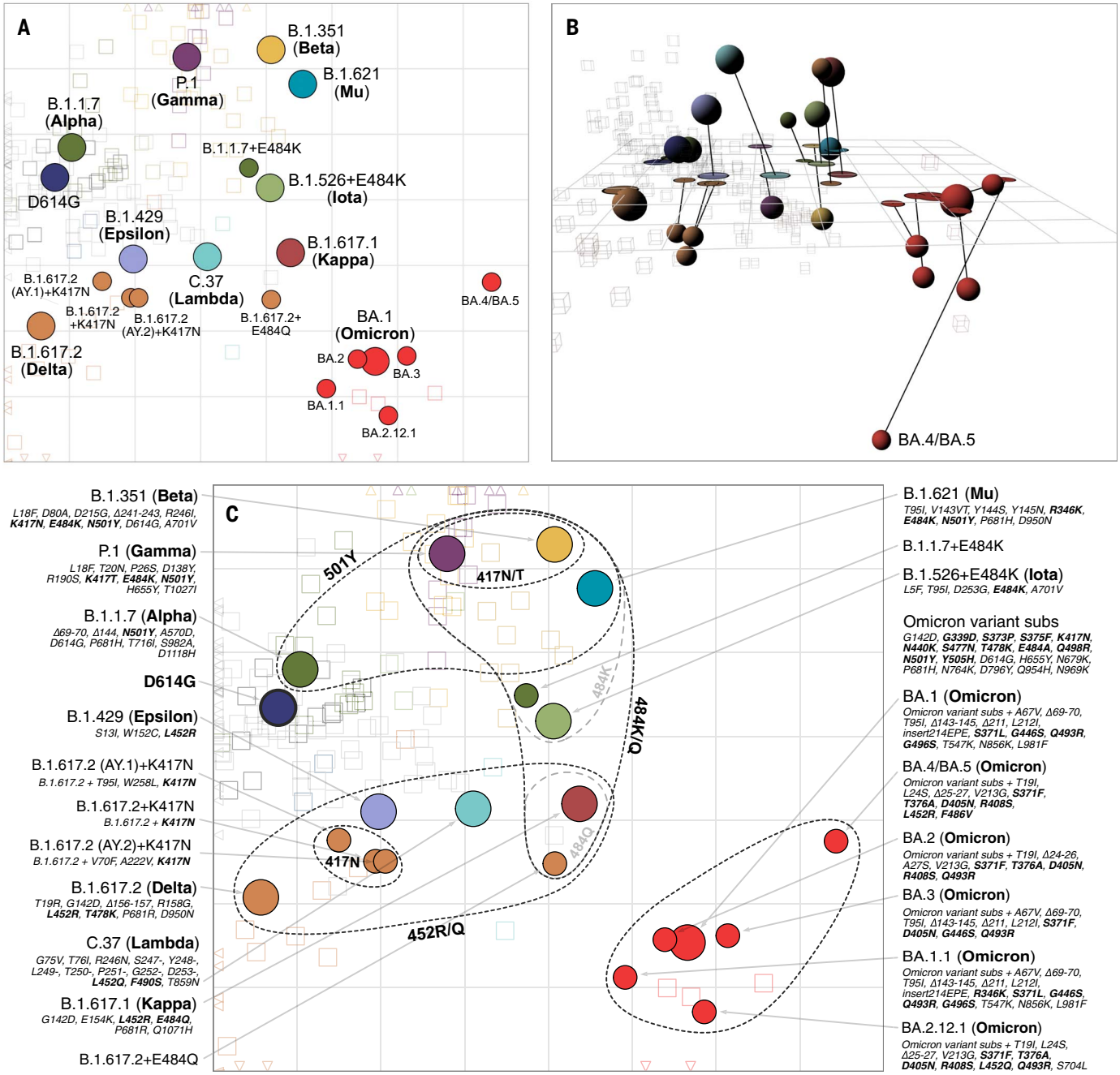

Fig. 3. Antigenic map of SARS-CoV-2 variants and selected substitutions. (**A**) Antigenic map with variant names. (**B**) Antigenic map with variant positions in 3D representation and lines connecting to their respective positions in the 2D map. (**C**) Antigenic map with variant names and substitutions annotated and grouped by amino acid present at spike positions 417, 452, 484, and 501, with an additional grouping for the six Omicron variants. Variants are shown as circles, sera as squares. Variants with additional substitutions from a root variant are denoted by smaller circles in the color of their root variant. The *x* and *y* axes both represent antigenic distance, with one grid square corresponding to a twofold serum dilution in the neutralization assay. Therefore, two grid squares correspond to a fourfold dilution, three to an eightfold dilution, and so on. The *x*-*y* orientation of the map is free because only the relative distances between variants and sera are relevant. Triangular arrowheads at the edge of the bounding box point in the direction of the sera that would be shown outside of the plot limits. A nonzoomed version of this map is shown in fig. S24. Interactive versions of the maps shown in (A) and (B) are available at https://acorg.github.io/mapping_SARS-CoV-2_antigenic_relationships_and_serological_responses.

BA.1 sera or to inherent asymmetries in BA.1 and pre-Omicron sera cross-reactivity. In general, the result is consistent with other results showing that BA.1 infections generate low titers to pre-Omicron variants (*29*, *30*).

The landscapes for different mRNA-1273 postvaccination sera confirm expectations for how cross-reactivity differs depending on the number of, and time since, vaccinations. These differences can largely be modeled by different slopes of titer reduction across antigenic space, with reactivity that peaks in the same antigenic region but decreases at differing rates (Fig. 4, K to M, and Fig. 2, B and C). The breadth of increase is clear at >3 months after a second

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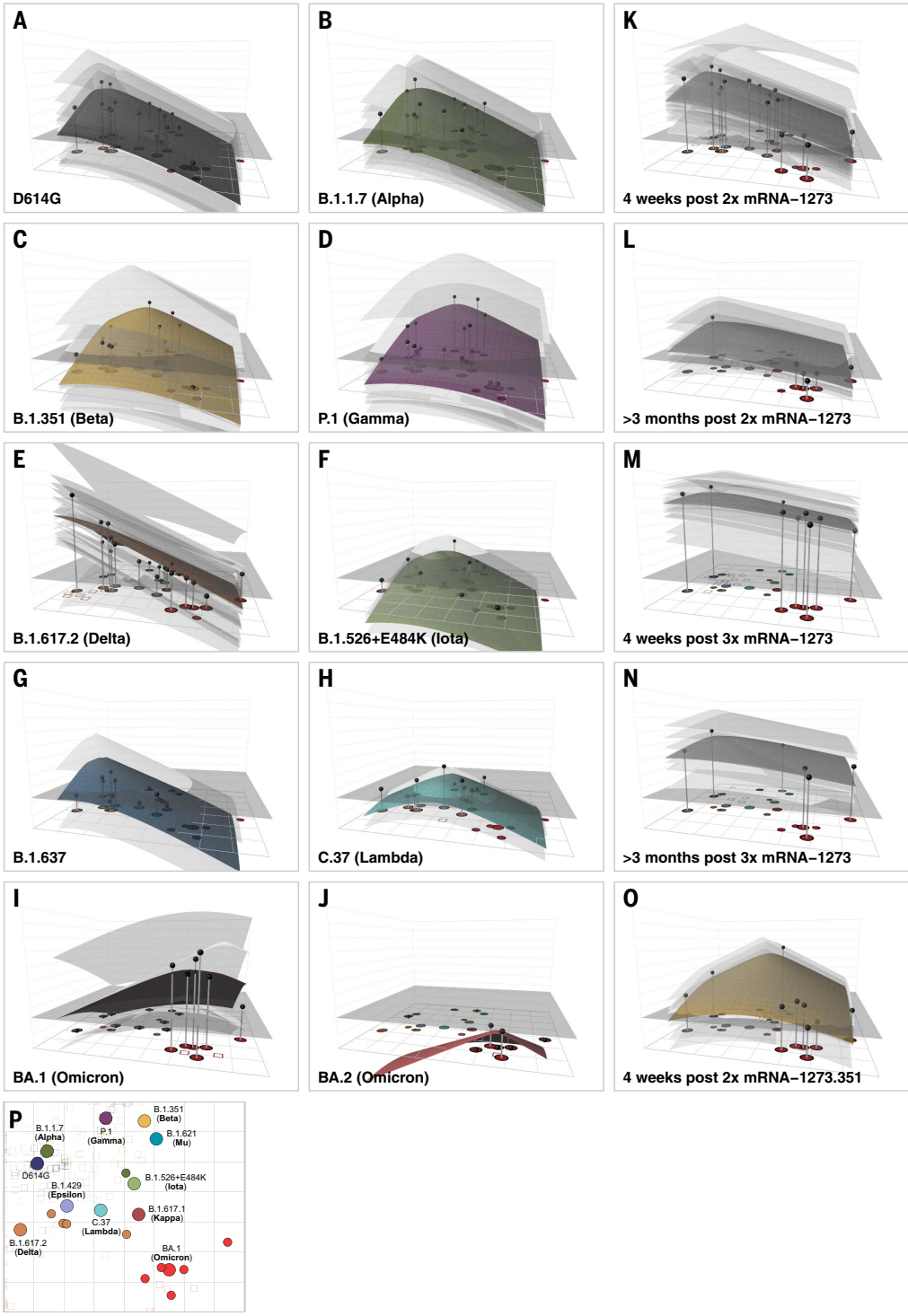

Fig. 4. Antibody landscapes for each serum group. Colored surfaces show the GMT antibody landscapes for the different serum groups; light-gray surfaces show the landscapes for individual sera. Gray impulses (vertical lines) show the height of the GMT for a specific variant, after accounting for individual effects as described in Materials and methods (which would otherwise bias the GMT for variants not titrated against all sera). The base *x-y* plane and plane colors correspond to the antigenic map shown in Fig. 3 and reproduced in (**P**). The vertical *z* axis in each plot corresponds to the titer on the $\log_2$ scale; each twofold increment is marked, starting from a titer of 20, one unit above the map surface. The gray horizontal plane indicates the height of a titer of 50 as a reference for judging the landscapes against various estimates of neutralizing antibody correlates of protection. Additional visualizations of predicted versus fitted titers are shown in fig. S26. The number of sera included for the calculation of the landscapes are (**A**) D614G sera (n = 13), (**B**) B.1.1.7 sera (n = 13), (**C**) B.1.351 sera (n = 15), (**D**) P.1 sera (n = 13), (**E**) B.1.617.2 sera (n = 21), (**F**) B.1.526+E484K sera (n = 4), (**G**) B.1.637 sera (n = 2), (**H**) C.37 sera (n = 2), (**I**) BA.1 sera (n = 4), (**J**) BA.2 sera (n = 1), (**K**) 4-weeks post–2× mRNA-1273 sera (n = 30), (**L**) >3-months post–2× mRNA-1273 sera (n = 13), (**M**) 4-weeks post–3× mRNA-1273 sera (n = 26), (**N**) >3-months post–3× mRNA-1273 sera (n = 8), and (**O**) 4-weeks post–2× mRNA-1273.351 sera (n = 9). Interactive versions of the landscapes in each of the panels are available at https://acorg.github.io/mapping_SARS-CoV-2_antigenic_relationships_and_serological_responses.

mRNA-1273 vaccine dose and changes magnitude but maintains similar breadth at 4 weeks and >3 months after a third mRNA-1273 vaccine dose.

Molecular basis of the antigenic map topology

As shown in Fig. 3C, the locations of variants in the antigenic map point to amino acid substitutions that are shared between pre-Omicron variants with similar antigenic characteristics. For example, variants with substitutions at position 484 (E to K or Q) are positioned on the right of the map because of poorer neutralization by D614G and vaccine sera. Variants toward the top of the map [B.1.1.7, B.1.351, P.1, B.1.621 (Mu), and B.1.1.7+E484K] all have a substitution at position 501 (N to Y). Variants B.1.351 and P.1 additionally have substitutions at position 417, indicating that these changes are associated with increased reactivity to B.1.351 and P.1 sera. Variants in the lower half of the map [B.1.617.2, B.1.429 (Epsilon), C.37, and B.1.617.1 (Kappa)] all have substitutions at position 452 (L to R or Q). The Omicron variants, carrying at least 15 additional substitutions in the RBD, form a separate cluster in the lower right of the antigenic map.

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To further investigate the molecular basis of these antigenic differences, we generated 10 lentivirus pseudotypes with single substitutions at positions 417, 452, 484, and 501 in different RBD contexts and measured the effect on reactivity to different serum groups and the subsequent positioning of variants in the antigenic map. As shown in Fig. 5, the antigenic effect of the different substitutions when introduced in isolation was generally consistent with that inferred from the antigenic map of wild-type variants. Large antigenic effects were seen for substitutions at position 484, which were associated with the right-left antigenic variation seen in the map, such that D614G+E484K showed greater escape than D614G+E484Q in D614G sera (Fig. 5, A and B, and fig. S27, A and B). Smaller but significant effects were seen for substitutions at position 417, which was shown to mediate some of the top-bottom map variation (Fig. 5B and fig. S28). Introduction of the N501Y substitution alone into D614G did not mediate large antigenic changes in the map but did cause significantly increased reactivity to B.1.1.7, P.1, and B.1.351 sera (fig. S29) and generated a virus that was antigenically similar to B.1.1.7, which also contains only the substitution N501Y in the RBD (Fig. 5A and fig. S1).

Despite overall correspondence with map-based predictions, some results were unexpected. In particular, although D614G+L452R showed decreased reactivity to D614G sera (fig. S27C), the D614G+L452R+E484Q mutant did not show any significant difference in reactivity compared with that of the D614G+E484Q mutant (fig. S27, B and D). Moreover, the B.1.429+K417N mutant showed increased reactivity to the D614G sera (fig. S28D), even though this represents a change away from sequence identity with D614G. Because it has been shown that the presence of the K417N substitution alone in the absence of 501Y greatly reduces angiotensin-converting enzyme 2 (ACE2) affinity in some contexts (*31*), we speculate that the effect seen in B.1.429+K417N may be influenced by an artificial inflation of titers generally.

Furthermore, the introduction of the A484K substitution into the BA.1 context allowed us to compare the effect of the alanine (A) substitution present in the Omicron variants in contrast to the lysine (K) substitution seen in pre-Omicron antigenic escape variants, such as B.1.351. We found that the BA.1+A484K mutant had lower titers compared with the BA.1 variant for sera taken 4 weeks after a second mRNA-1273 vaccine dose and also for B.1.617.2 sera (fig. S30), raising the question as to why this alternative substitution has not been seen more frequently in Omicron variants in nature. In terms of movement in the antigenic map, in two dimensions, the BA.1+A484K mutant moves to a location to the top right of the BA.1 variant, bringing it closer to the B.1.351 sera but also consequently close to BA.4/BA.5 (fig. S31). However, antigenic relations with BA.4/BA.5 are better described in 3D representation (Fig. 3B), in which the BA.4/BA.5 variant uses the third dimension and the BA.1+A484K mutant occupies a novel area of antigenic space distinct from the other Omicron variants (Fig. 5C).

Variation in immunodominance of different RBD sites between serum groups

Overall, among the substitutions tested, not all serum groups were equally sensitive to changes at a given position, with some substitutions having a large effect on reactivity to certain serum groups but little to no effect in others. For example, although the D614G+E484K and D614G+E484Q substitutions had large effects on reactivity to D614G sera, no significant effect on B.1.351 serum reactivity was found (fig. S27, A and B). Such findings are consistent with variation in immunodominance patterns and in the extent to which antibodies in different sera target different structural regions in the RBD.

We sought additional evidence for immunodominance switches by analyzing results from the mutants alongside differences in serum reactivity between other pairs of variants that differed by single–amino acid substitutions in the RBD (Fig. 6A and figs. S32 and S33).

For all pairs of variants with differences in the RBD only at position 484 from E to K or Q (Fig. 6A, rows 1 and 2), D614G sera consistently showed significantly reduced reactivity [by using the 95% highest-density interval (HDI) estimate of the pairwise difference in titers as a criterion throughout], whereas the B.1.351 sera showed little to no significant change in reactivity. Across the other serum groups, sera from individuals infected with variants with the ancestral 484E (2× mRNA-1273, D614G, B.1.1.7, and B.1.617.2) were generally sensitive to differences at position 484, whereas sera from infections with variants with 484K (B.1.351, P.1, and B.1.526+E484K) were more resistant to these substitutions, although B.1.351 did show evidence for some smaller increases in titers in some cases. The BA.1+A484K mutant also reflected this pattern of sensitivities in the serum groups against which it was titrated, with significant titer decreases in mRNA-1273 and B.1.617.2 sera (484E) and a smaller increase in reactivity to mRNA-1273.351 sera (484K) (fig. S30, S33). Of the four BA.1 sera (484A), the mean fold decrease of 2.1 against the BA.1+A484K mutant versus the BA.1 variant indicated that these sera are also somewhat sensitive to changes at the 484 position. However, this effect was far less than that seen for the D614G+E484K mutant on reactivity to post-D614G infection

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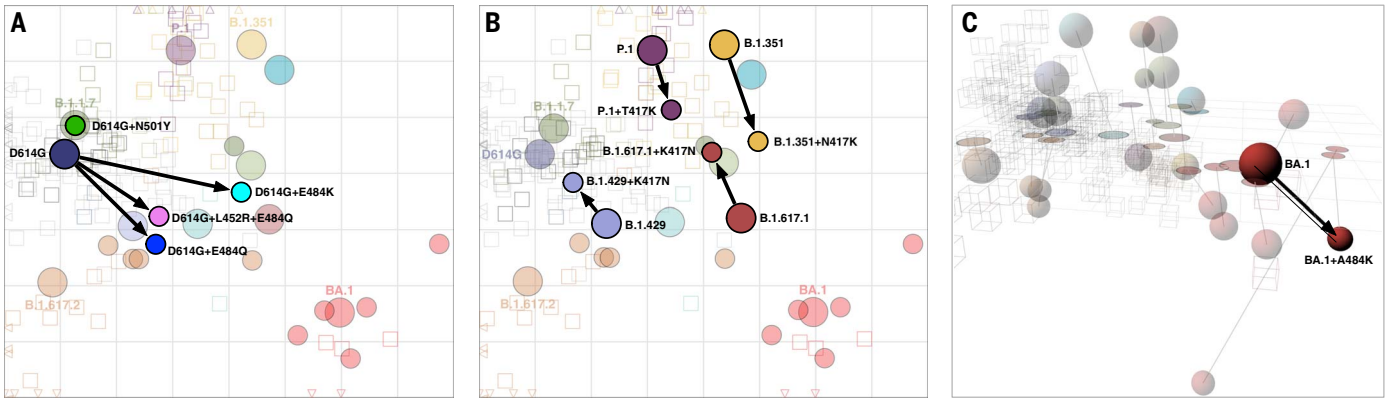

Fig. 5. Antigenic maps including laboratory-made mutants with substitutions at positions 417, 452, 484, and 501. (**A**) Variants with substitutions N501Y, E484K, E484Q, and L452R+E484K in the background of D614G; D614G+L452R is not shown because it was titrated against only D614G sera, so its position could not be determined. (**B**) Variants with the T/N417K substitution in the background of P.1 and B.1.351, respectively, and K417N in the background of B.1.429 and B.1.617.1. (**C**) BA.1 with the substitution A484K. The map in (C) is 3D to highlight the antigenic differences between BA.1, BA.1+A484K, and BA.4/BA.5. The 2D version of (C) is shown in fig. S31. Arrows point from the antigenic position of the root virus to that of the laboratory-generated variant. Interactive versions of the maps shown in each panel are available at https://acorg.github.io/mapping_SARS-CoV-2_antigenic_relationships_and_serological_responses.

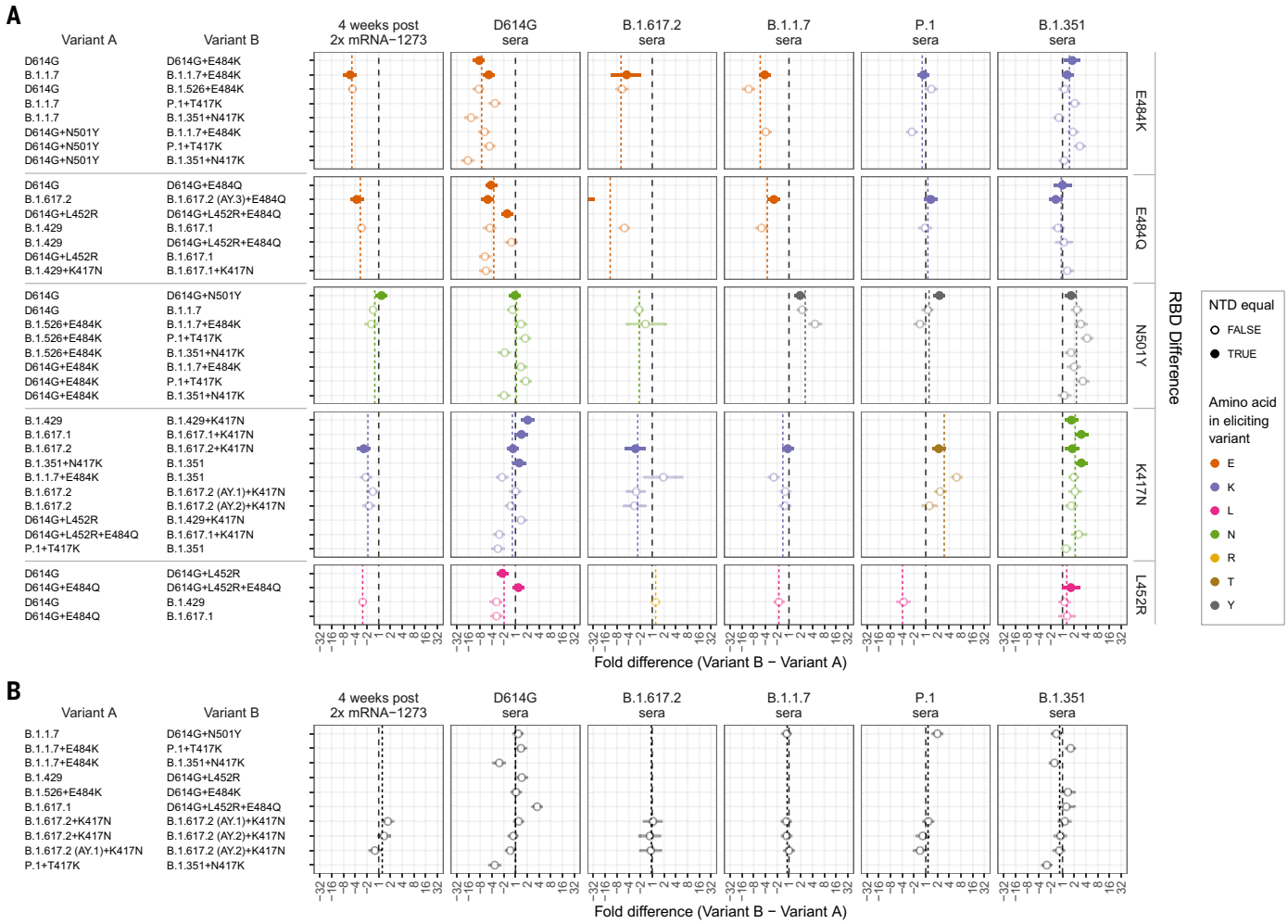

Fig. 6. Effect of pairwise amino acid differences on reactivity to different serum groups. This plot compares the average fold difference in titer between (**A**) different pairs of variants that differ by only a single amino acid difference in the RBD or (**B**) that do not differ by any amino acids in the RBD but differ in the NTD. Comparisons are grouped by serum group (panel columns) and corresponding RBD difference (panel rows). In each panel, the circle represents the estimate for the average fold difference in titer between variant A and variant B, as named on the left-hand side of the plot, and the lines extend to indicate 95% HDI for this estimate. The black dashed line marks a fold difference in titer of 1 (no difference), and the colored dashed line indicates the average fold difference between all pairs of variants with that substitution in the RBD. Points and lines are colored according to the amino acid in the variant homologous to that serum group, at the position in the RBD where the pair of variants compared differ. Filled circles indicate where pairs of variants had no additional amino differences in the NTD region, often because one was generated as an artificial mutant. By contrast, open circles indicate pairs of variants with additional amino acid differences in the NTD region, in addition to the RBD amino acid difference listed. The estimate for B.1.617.2 sera fold differences between the B.1.617.2 and B.1.617.2 (AY.3)+E484Q variants [(A), third column, second row], which falls outside the plot, is −59.4 (95% HDI −117.1, −30.7). Details of how fold-difference estimates and HDIs were calculated are described in Materials and methods. Figures S30 and S31 show the same results against an expanded set of pairwise amino acid differences. Interactive scatterplots comparing titers against each pair of variants for each serum group are available at https://acorg.github.io/mapping_SARS-CoV-2_antigenic_relationships_and_serological_responses.

and vaccination sera, suggesting that like the B.1.351 and P.1 serum responses, the BA.1 response also appeared to be directed away from the 484 position.

Where variants differed only by N501Y (Fig. 6A, row 3), we also found that sensitivity was linked to the amino acid at position 501 in the infecting variant. The two serum groups resulting from infections with B.1.351 and B.1.1.7 (both with 501Y) were sensitive to differences at position 501, whereas the serum groups postprototype vaccination and post-D614G, B.1.526+E484K, and B.1.617.2 infection (variants with ancestral 501N) were typically not sensitive to the N501Y substitution.

For K417N comparisons (Fig. 6A, row 4), there was evidence for equal or decreased titers in serum groups with exposure to variants with the ancestral K at position 417 (mRNA-1273, D614G, B.1.617.2, and B.1.1.7 serum groups) and a corresponding overall increase in titers to the serum group after infection with B.1.351 (417N). P.1 sera (417T) showed increased reactivity to the K417N substitution. The B.1.351 sera (417N) also showed increased reactivity associated with the K417T substitution, likely reflecting a structural homology of the changes caused by K417T and K417N.

In addition, where variants allowed for a comparison of the effect of the substitution L452R (Fig. 6A, row 5), we found evidence for most of the serum groups distinguishing between variants that differed by this substitution, with decreased (mRNA-1273, D614G, B.1.1.7, and P.1 sera) or increased (B.1.617.2 sera) titers, corresponding to the amino acid present at position 452 in the eliciting variant. One exception was the B.1.351 sera, in which, unlike the other serum groups with the ancestral 452L,

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the L452R substitution did not produce an overall decrease in titers.

Impact of changes in the NTD

We also investigated the effects of substitutions in the N-terminal domain (NTD). Generally, the effects of single–amino acid differences in the RBD were consistent regardless of whether additional NTD differences were present or not (Fig. 6A). This was also reflected in pairwise comparisons of variants with no RBD differences, in which we typically found no significant differences in titer reactivity (Fig. 6B). This included little evidence for a significant effect on serum reactivity of features, such as the NTD 69 to 70 deletion in the B.1.1.7 variant (Fig. 6B, B.1.1.7 versus D614G+N501Y). However, for comparators involving the B.1.351+N417K mutant, which has the substantial B.1.351 NTD changes that include the 241 to 243 deletion and R246I substitution, we did find evidence of differences compared with other viruses that had sequence identity in the RBD, such as the P.1+T417K and B.1.1.7+E484K mutants. Although these differences may be associated with the effects of these NTD substitutions on titer reactivity [as has been found in other cases (*32*)], we found that in the comparisons shown in Fig. 6B, the B.1.351+N417K mutant also had lower titers against the B.1.351 sera themselves, despite sequence identity of the NTD differences for these sera. This indicates that in our data, there was a general negative bias in titers measured against the B.1.351+N417K mutant, rather than antigenic effects of the substitutions within the NTD. Others have found that sensitivity to NTD substitutions can depend on the levels of ACE2 expression in the assay target cell line (*33*), which may relate to us finding little evidence overall for NTD effects in the cases that we compared.

Discussion

The antigenic analyses of SARS-CoV-2 variants presented here underscore the advantages of an integrated and extensible framework for understanding antigenic relationships and serum responses in SARS-CoV-2. The antigenic map and the antibody landscapes allow the comparison of serum responses by variant and also provide a quantitative measurement of both the response magnitude and breadth after different exposures. Antibody landscapes can also be applied to analyze how multiexposure serum responses distribute across antigenic space (*34*), which is important for ongoing studies seeking to compare the immunity built through different prospective vaccination regimens (*35*, *36*).

Using regression analyses and antibody landscapes to analyze immune responses in vaccination sera taken at multiple time points after vaccination, we quantified and visualized changes in response cross-reactivity breadth, disentangling it from changes in cross-reactivity that are due to differences in response magnitude (Fig. 2). The similar estimates of response breadth just before the third vaccination, and from 4 weeks after the third vaccination, showed that the main short-term effect of the third vaccination was to boost the magnitude of a response that had already become more cross-reactive, rather than to generate additional breadth of cross-reactivity. This is consistent with studies into influenza vaccine responses, in which the predominant effect of vaccination is boosting of preexisting patterns of prevaccination reactivity, which was independent of the vaccine variant used (*34*). It is possible that the third vaccine dose was responsible for the later additional increases in cross-reactivity in >3 months after third vaccine dose samples, although the effect size was small and not statistically significant. It is also possible that our observation of increasing response breadth over time (Fig. 2B) would have continued to this point without a third vaccination. However, increases in breadth in the absence of a third dose would not have translated to increased protection without the effect of the third dose boosting antibody titers.

Different variants and serum responses clustered on the antigenic map in a way that allowed the inference of candidates for the amino acid substitutions responsible for the main antigenic changes. The impact of substitutions at position 484, 417, and 501 agree with other work based on deep mutational scanning (DMS) (*37*–*39*) and neutralization data from vesicular stomatitis virus pseudotyped particles (*40*, *41*). The large antigenic effect of single substitutions such as E484K is reminiscent of influenza viruses, in which a large portion of the antigenic difference is associated with only a single or double substitution located adjacent to the receptor binding site on the hemagglutinin protein (*42*–*44*). This is also observed for the determinants of SARS-CoV-2 antigenic evolution described here, which are located close to the ACE2 binding site on the spike protein (figs. S34 and S35).

Variation in immunodominance among distinct antigenic regions is gaining recognition as an important aspect of the antigenic evolution of pathogens and has also been seen in influenza, hepatitis C, and noroviruses (*45*–*47*). Conversely, the codominance of multiple sites in measles virus has been associated with reduced potential for evasion from serological reactivity (*48*). Within SARS-CoV-2, our findings are consistent with studies showing that monoclonal antibodies tend to have different regional binding preferences when sourced from individuals exposed to different variants (*10*, *49*–*51*). Conclusions suggesting an immunodominance switch for B.1.351 sera away from the 484 region also correspond with published data applying DMS to sera after D614G and B.1.351 infection (*39*). Our results show how such immunodominance switches extend to vaccination and infections with different variants and how changes can be associated with the amino acid present in the different eliciting variants. These observations explain the wider patterns in the data—in which certain serum groups may distinguish clear antigenic differences between certain variants although others may not—and again underscore that antigenic differences between variants are not necessarily absolute measures but can be relative to the particular sera against which variants were titrated and the specific structural regions that the particular sera tested predominantly target. Immunodominance switches also highlight the potential limitations of visualizing antigenic relationships solely in Euclidean space, as is done with antigenic maps, and emphasize the need for further complementary approaches to their quantification and visualization, such as the explicit fold-drop comparisons that we use here.

We note two limitations in the analyses that we have presented. First, although the assay that we use is FDA certified, it measures neutralization with a lentivirus pseudotype neutralization assay, whose results may differ from those of live-virus neutralization assays. However, we generally find our fold-difference estimates to be within the ranges reported in other studies and, in agreement with others (*12*), we do not find an obvious bias compared with live-virus neutralization assay results (fig. S7). Second, we focus on patterns of serological reactivity and antigenic variation found up until the early Omicron variants. Beyond this point, it is very difficult to source sufficient primary-exposure human sera to study antigenic relationships in detail, and it is increasingly necessary to rely on animal-model sera. Consequently, it is critical to assess whether the patterns of antigenic relatedness and changes in immunodominance found for human sera can be reliably measured in different animal models.

As populations increasingly experience multiple exposures, it will be important to investigate how antigenic differences in multiexposure serum responses compare with those inferred from primary-exposure sera, in particular with immunodominance patterns in mind. For example, findings that infection with Omicron BA.1 after a previous exposure to pre-Omicron variants predominately boosts antibodies targeting epitopes shared between pre-Omicron and Omicron variants highlight the value of understanding how different initial exposures dictate the structural regions that were initially targeted and how this interacts with a future exposure (*52*–*55*). Answers to these questions will better reveal the variants and substitutions to which different populations are most vulnerable and help anticipate which emerging variants may be most at risk of evading

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current immunity. The choice of vaccine immunogens on the basis of immunodominance considerations may become as important as their antigenic characteristics.

Materials and methods

Specimens and study cohorts

All samples, including those from vaccine recipients, were collected in clinical studies with information listed in table S1. Convalescent sera are assumed but not confirmed first infections, based on no previously recorded or self-reported SARS-CoV-2 vaccination or infection, plus additional criteria in some cases, as also described in table S1. For convalescent samples, the infecting variant was determined using whole-genome sequencing. Institutional review board (IRB) approvals were obtained for each study site. All participants were given and signed the informed consent form.

Vaccination samples came from a variety of sources, as also indicated in table S1 and described in more detail below:

4 weeks post 2× mRNA-1273

The majority (n = 28) of the 4-weeks post–2× mRNA-1273 samples were from the mRNA-1273 phase 1 study (NCT04283461) and were obtained from the Division of Microbiology and Infectious Diseases (DMID), National Institute of Allergy and Infectious Diseases (NIAID) for the mRNA-1273 phase 1 study team, and Moderna. The phase 1 study protocols and results are reported previously (*57*, *58*). The phase 1 trial tested the identical vaccine (mRNA-1273), dose (100 μg), and schedule as used in the Moderna phase 3 (NCT04470427); titrations against these samples were also reported in (*59*). A smaller subset of samples (n = 4) were obtained from the Coronavirus Efficacy (COVE) phase 3 trial and were preselected as having high titers of neutralizing antibodies against D614G to enable detection of neutralization against the wider range of Omicron variants that they were measured against; some titrations against these samples also appear in (*60*). Finally, an additional 11 sera from the COVE phase 3 trial were titrated against D614G and the D614G+N501Y mutant, and an additional 26 sera were titrated against BA.1 and the BA.1+A484K mutant. These sera appear in the results shown in fig. S29 and S30, respectively, but are not included in the general analysis due to lack of titrations against further variants.

>3 months post 2× mRNA-1273

All samples relate to sera previously described in (*61*) that were obtained before homologous mRNA-1273 boosting (50 μg) in 16 study participants who had received two inoculations of mRNA-1273 (100 μg) under emergency-use authorization. These participants were part of a subset analysis of high D614G responders in the NIAID heterologous boost study DMID 21-0012.

4 weeks post 3× mRNA-1273

Sixteen samples relate to sera from the same individuals in the >3-months post 2× mRNA-1273 group, but obtained 29 days after homologous mRNA-1273 boosting (50 μg), again as described and appearing in (*61*). An additional 10 samples also relate to samples collected as part of the DMID 21-0012 and also relate to samples previously described in (*60*), which were also a subset of high D614G responders.

>3 months post 3× mRNA-1273

These samples were taken as part of the COVID-19 Variant Immunologic Landscape (COVAIL) randomized clinical trial (NCT 05289037) and are from the subset of individuals who had received a previous mRNA-1273 vaccine regimen followed by a mRNA-1273 booster dose. Samples were taken directly prior to a fourth mRNA-1273 dose and, as for other vaccine groups, we subset to those with no previous record of a SARS-CoV-2 infection.

4 weeks post 2× mRNA-1273.351

Samples were from COVID-19 naïve participants who received two 100-μg doses of mRNA-1273.351 on days 1 and 29 as part of a clinical trial (NCT 04785144), experimental arm 2C.

The use of the specimens for the research conducted in this manuscript is covered under IRB protocols Pro00093087 and Pro00105358.

Where samples in the same serum group came from a mixture of cohorts or agencies, we did not find evidence for differences in the patterns of reactivity according to cohort or agency (fig. S3).

Pseudotype virus assay

Neutralization was measured with lentiviral particles pseudotyped with SARS-CoV-2 spike and containing a firefly luciferase (Luc) reporter gene for quantitative measurements of infection by relative luminescence units (RLU). The assay was performed in 293T/ACE2-MF provided by Michael Farzan (-MF) and Huihui Mu. The derivation of the cell line is described in (*62*). Pseudoviruses were prepared, titrated, and used for measurements of neutralizing antibodies essentially as described previously (*59*). Briefly, an expression plasmid–encoding codon-optimized full-length spike of the Wuhan-1 ancestral sequence (VRC7480) was provided by Barney Graham and Kizzmekia Corbett at the Vaccine Research Center, National Institutes of Health (USA). Mutations were introduced into VRC7480 either by site-directed mutagenesis using the QuikChange Lightning Site-Directed Mutagenesis Kit from Agilent Technologies (Catalog no. 210518) or were created with a spike gene synthesized by GenScript, using the spike sequence in VRC7480 as template. All mutations were confirmed with full-length spike-gene sequencing by Sanger Sequencing, using Sequencher and SnapGene for sequence analyses. Pseudovirions were produced in HEK293T/17 cells (ATCC cat. no. CRL-11268) by transfection using Fugene 6 (Promega Cat no. E2692) and a combination of spike plasmid, lentiviral backbone plasmid (pCMV ΔR8.2), and firefly Luc reporter gene plasmid (pHR′ CMV Luc)36 in a 1:17:17 ratio in Opti-MEM (Life Technologies) (CMV, cytomegalovirus). Transfection mixtures were added to preseeded HEK 293T/17 cells in T-75 flasks containing 12 ml of growth medium and incubated for 16 to 20 hours at 37°C. Medium was removed and 15 ml of fresh growth medium added. Pseudovirus-containing culture medium was collected after an additional 2 days of incubation and clarified of cells by low-speed centrifugation and 0.45-μm micron filtration. Median tissue culture infectious dose ($TCID_{50}$) assays were performed as described previously (*59*).

For measurements of neutralization, a pretitrated dose of virus was incubated with eight serial fivefold dilutions of serum samples (1:10 or 1:20 starting dilution) in duplicate in a total volume of 150 μl for 1 hour at 37°C in 96-well flat-bottom culture plates. 293T/ACE2-MF cells were detached from T-75 culture flasks using TrypLE Select Enzyme solution, suspended in growth medium (100,000 cells/ml) and immediately added to all wells (10,000 cells in 100 μl of growth medium per well). One set of eight wells received cells and virus (virus control), and another set of eight wells received cells only (background control). After 66 to 72 hours of incubation, medium was removed by gentle aspiration, and 30 μl of Promega 1X lysis buffer was added to all wells. After a 10-min incubation at room temperature, 100 μl of Bright-Glo luciferase reagent was added to all wells. After 1 to 2 min, 110 μl of the cell lysate was transferred to a black or white plate. Luminescence was measured using a GloMax Navigator luminometer (Promega). Serum samples were heat-inactivated for 30 min at 56°C prior to assay. Neutralization titers are the inhibitory dilution (ID) of serum samples at which RLUs were reduced by either 50% (ID_{50}) or 80% (ID_{80}) compared with virus control wells after subtraction of background RLUs. The spike mutations of all variants used in this study are shown in table S2.

Titer analyses

Excluding outlier sera

The serum samples used in this study were collected from individuals with no reported previous vaccination or infection preceding the exposure that was associated with their respective serum group, with further additional criteria for some sample sources as described in table S1. However, as with most studies, it is possible that for some individuals, additional previous (even asymptomatic)

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infections were not detected or reported. Based on this possibility, we sought to identify sera with patterns of reactivity that were less likely to be consistent with a true first exposure. As such, we applied selection criteria to all the sera to identify cases where a titer was measured that was higher than the serum-group-homologous-variant titer by more than twofold (for example, a serum from the B.1.617.2 group that had a titer against D614G that was more than twofold higher than the titer measured against B.1.617.2). Subvariants of B.1.617.2 and BA.1 (Fig. 3, smaller circles) were excluded from this analysis, as well as the variant P.1, which we noted tended to often have high titers generally. For example, P.1 often has higher-than-homologous titers for sera from the B.1.351 serum group, possibly due to assay-related biases inflating titers for the particular P.1 variant, as it has been observed in other assays using lentiviral pseudotypes (9). B.1.429 was used as the homologous variant for sera from the B.1.637 group, being the genetically most similar variant titrated, while D614G was used for mRNA-1273 vaccination groups; and B.1.351 was used for the mRNA-1273.351 vaccination group. Fifteen of the total of 207 sera had titers exceeding the twofold-higher-than-homologous criteria described above and are highlighted in red in the titer plot shown in fig. S2.

As an additional step, we also excluded nine sera as likely second-infection responses based on visual inspection of patterns of reactivity that deviated strongly from the general trend of titer reactivity for a given serum group. These sera are additionally highlighted in fig. S2 in blue. In particular, this criteria applied to sera from the BA.1 serum group where we noted a clearly bimodal pattern of reactivity with three post-BA.1 infection sera that showed high titers against the D614G variant and four sera which did not. Based on the large antigenic difference between D614G and BA.1 and the types of reactivity seen in more controlled animal-model BA.1 infections (63), we judged the four more-BA.1-specific sera to be likely first infections and excluded the rest of the group as likely cases that had had a previous exposure to a pre-Omicron variant.

Calculation of Geometric Mean Titers (GMT) and fold changes for nondetectable titers

Calculation of the mean titers becomes nontrivial when experiments contain nondetectable titers or, in data science terminology, censored data. Although computation of means of data with censored values has been around for over 60 years (64–66), it does not seem to be commonly employed when analyzing data from neutralization assays. Correct estimation of the GMT becomes especially important if one wants to quantify the amount of escape by variants like the Omicron variants, with many nondetectable titers. Setting nondetectable titers to half the value of an assay's limit of detection is a common method employed in virology, however, it may overestimate the GMTs for such variants. Below we give a brief overview of our approach to dealing with such censored data, functions for which are included in the associated titertools package (67).

The main step in estimation of the titers is the construction of a likelihood function that describes the probability of observing both the nondetectable and detectable titers. In the dataset used for this study, we deal with only left-censored data. We assume that $\log_2$ titer values of a particular serum group are modeled by a normal density function, $f(x;\mu,\sigma)$, with unknown mean μ and standard deviation (SD) σ, and we let $y_i (i = 1, ..., M)$ denote detectable titers and $T_i (i = 1, ..., N)$ denote the nondetectable titers. Then the log likelihood function, LL, is given by

$$\mathrm{LL}(\mu,\sigma) = \sum_{i=1}^{N} \log \int_{-\infty}^{T_i} f(x;\mu,\sigma) + \sum_{i=1}^{M} \log f(y_i;\mu,\sigma)$$

In this expression, the term involving the integral is the likelihood of observing the censored titers, whereas the other term is the likelihood of observing the noncensored titers. This likelihood function then can either be used in a maximum-likelihood estimation pipeline or, if priors are chosen for μ and σ, in a Bayesian posterior-calculation pipeline from which the value of μ can be estimated in various ways. For the calculations in this paper, confidence intervals were calculated using the highest density interval (HDI) method of calculation from 10,000 samples.

When calculating the fold changes in titers, the same approach to nondetectable titers was taken, with, for example, a change from 40 to <20 equating to a change of <−1 on the $\log_2$ scale, for which the likelihood function would be the integral of the normal density function from $-\infty$ to −1. For estimating fold changes, we used the function "log2diff" in the titertools package, with parameter "sigma = 0.87," based on our estimates of the magnitude of variation in paired measurement repeats (fig. S10).

Titers after accounting for estimated individual effects

The data used in this paper contain sera collected from different individuals infected with the same variants. Even when these titers are grouped according to the infecting variant, one can see quite large variations in titer magnitudes, whereas the qualitative appearance of the titer curve is similar (Fig. 1). Some of this is due to individual variation in the overall reactivity of sera collected from different individuals, for example, related to the magnitude of the serological response. We model this behavior in the following way: Assume that t_{ij} is the $\log_2$ titer measurement of antigen i against serum j from the serum group J. Then

$$t_{ij} = s_{iJ} + r_j + \varepsilon_{ij} (i = 1, ...n \text{ and } j = 1, ...m)$$

where s_{iJ} is the average $\log_2$ titer of antigen i against the serum group J, r_j is the serum reactivity bias of the serum j, and ε_{ij} is the independently and normally distributed log-titer noise. n is the number of antigens, and m is the number of sera. For each serum group, parameters s_{iJ} and r_j are then found to maximize the sum of the likelihood of each measured log titer given the observed log titer, t_{ij}. For the fitting procedure, we used the function "estimate_sr_effects" in the titertools package, with parameter "sigma = 0.62," according to our estimates of the magnitude of measurement noise in a single measurement (fig. S10).

Antigenic cartography has its own method of dealing with serum reactivity bias. The formula for the target distance (provided in "Antigenic cartography" below) indicates that for any serum, the antigen with the highest titer (usually the homologous antigen, but not always) is automatically adjusted to have 0 distance to that serum. This is true regardless of the value of the maximum titer, and this adjustment also takes care of the individual serum reactivity in a way that is compatible with antigenic mapmaking.

Calculating fold-drop differences in vaccine sera

When exploring differences between postinfection and different postvaccination responses homologous to the same antigen, one might expect that the fold drop in serum responses (compared with the homologous antigen) are similar in some way. A simple model one can use is to introduce an overall factor that relates the responses of sera from the two groups. In terms of the model in the previous section, this means

$$s_{hJ} - s_{iJ} = d_J(s_{hK} - s_{iK}) = d_J \Delta s_{hi}{}^K$$

where K and J are two serum groups, h is the homologous antigen to them, i is any other antigen, and s_{xY} are as described in the previous section. Note that terms such as $s_{hJ} - s_{iJ}$ give the (noiseless) fold drop of titers of antigen i from the titers of the antigen h (homologous to J), measured against the serum group J. Then for a specific serum j from the serum group J, we have

$$\begin{aligned} t_{hj} - t_{ij} &= s_{hJ} - s_{iJ} + \varepsilon_{hj} - \varepsilon_{ij} \\ &= d_J \Delta s_{iK} + \varepsilon_{ijh} \end{aligned}$$

where the last term is again normally distributed however with $\sqrt{2}$ times the SD of the titer noise.

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An example would be K = D614G postinfection serum group, J = mRNA-1273 serum group, h = D614G antigen, and i = any other antigen (Fig. 2). The term d_J is the "factor" specific to serum group J, which relates its responses to the response of serum group K. (This term obviously also depends on K; however, we imagine that K is a constant postinfection serum group and that J varies between different postvaccination serum groups, so by abuse of notation, we only index this term with J.) Note that taking difference of titers cancels out the serum response bias specific to serum j.

When applying this model, we compared the serum groups D614G, 4 weeks post 2× mRNA-1273, >3 months post 2× mRNA-1273, and 4 weeks post 3× mRNA-1273 (with shared homologous variant D614G), and we compared serum groups B.1.351 and 2× mRNA-1273.351 (with shared homologous variant B.1.351). In the D614G and B.1.351 postinfection groups, we fixed d_J as 1, estimating fold-change differences in the postvaccination groups retive to the postinfection sera. For the fitting procedure, we used the function "compare_crossreactivity_breadth" in the titertools package, with confidence intervals calculated for the parameters based on the highest density interval of 5000 sampling runs after 1000 warmup runs.

Imputing censored titers

For figs. S11, S12, and S20, we imputed nondetectable titers in order to not bias the distributions that are visualized. In these cases, censored titers (e.g., <20) are imputed using the function "impute_gmt_titers" from the titertools package. In these cases, censored titers are replaced with numeric titers drawn from a censored normal distribution with mean and SD parameters that are estimated from the titer data as described in the "Calculation of GMT" section above.

Antigenic cartography

Antigenic cartography is a multidimensional scaling method (MDS) (68) for the reduction of dimensions in large-titer datasets in order to not only visualize large amounts of titer data in a lower-dimensional space but to also draw conclusions about the antigenic evolution of pathogens by studying the positions occupied by these viruses on the antigenic map.

In order to construct an antigenic map, one starts with a titer table where columns are sera and rows are antigens. Denoting the values of such a table by t_{ij} (where i denotes the antigen index and j the sera index), titer values represent a degree of dissimilarity of the ith antigen to the jth serum. We turn this dissimilarity score into a similarity score by the following transformation:

$$D_{ij} = \max_i \left[\log_2(t_{ij})\right] - \log_2(t_{ij})$$

The D_{ij} values are called the "target distances." Antigenic cartography strives to find a multidimensional (usually two or three dimensions) placement of antigens and sera, such that the Euclidean distance between antigens and sera conform as well as possible to the target distances. This is achieved by an optimization of the antigen and serum coordinates that minimize the stress function

$$\sum_{i,j} \left(D_{ij} - d_{ij}\right)^2$$

where d_{ij} represents the Euclidean distance between the ith antigen and the jth serum points in the antigenic map. The minimization is achieved using the L-BFGS algorithm (69) to find the best cost-minimizing coordinates starting from a set of randomly determined initial conditions. Note unlike most common MDS problems, the similarity matrix is not square and might contain masked or censored values and therefore requires a slightly custom approach. In particular, if t_{ij} is a nondetectable titer, then the stress is computed as

$$\varphi(D_{ij} - d_{ij} + s)(D_{ij} - d_{ij} + s)^2$$

where s is a step-size factor which is 1 for discrete data (and their means) such as those obtained from Haemagglutinin Inhibition assays, or 0 when the output data are continuous and measured from titer curves, and

$$\varphi(x) = \frac{1}{1 + 10^{-x}}$$

is a damping function whose role is to give more freedom to nondetectable titers to be optimized for a value less than the limit of detection of the assay.

We used the Racmacs package to compute the antigenic maps presented in this manuscript (70). The map was constructed using 1000 optimizations, dilution step size of 0, with the minimum column basis parameter set to "none." In order to allow for accurate coordination of serum positions in two dimensions, sera with less than three detectable titers were excluded from the maps. An exception was made for the BA.2 serum sample, which was included despite only having two detectable titers because it was the only serum of its group, although the serum position for this serum will therefore be less constrained in its antigenic position.

Assessing model fit

We performed different analyses to ascertain that the 2D antigenic map is a good representation of the titer data and to ensure that the map is robust to noise from measurement errors.

First, to get an understanding of the measurement error of individual titrations, we analyzed a set of repeat titrations of 60 sera against eight of the variants used in the assay and found that the mean difference in $\log_2$ titers between repeated titrations was 0.209 with a SD of 0.84, with measurement error being approximately normally distributed (fig. S10). Accounting for the systematic bias of titrations between repeats, and assuming a mean of 0 without it, resulted in a total SD of 0.87 on the $\log_2$ scale. Since measurement error would be present for both the first and second repeat titration, this would be consistent with noise per measurement with a SD of $\sqrt{0.87^2/2} = 0.62$.

Subsequently, we investigated how well antigenic maps of different dimensions represent the measured patterns of serological reactivity. In our previous experience with other antigenically variable pathogens, the dimensionality of relationships can be lower [for example, two dimensions in seasonal influenza and dengue virus (13, 20)] or higher [for example, three dimensions for A/H5N1 and swine influenza (17)] or may simply not be well represented in Euclidean space. Under cross-validation, a 2D map represented the data best (fig. S9A). The overall arrangement of the variants and sera in three dimensions is similar to that in two dimensions (fig. S9, B andC), with the exception of the BA.4/BA.5 variant, which in three dimensions occupies a position more distant from BA.1 and closer to B.1.617.2 and B.1.351 than visualized in two dimensions. The dimensional analysis was performed with Racmacs by using the function "dimensionTestMap."

Assessing goodness of fit using fitted and measured titers

Next, we assessed how well the 2D map fits the measured titers. We first investigated whether the measured $\log_2$ titers correlated with the titers fitted in the antigenic map. Overall, the fitted and measured titers corresponded well (fig. S11). When comparing the mean differences of the fitted versus measured titers (mean of 0.04 and SD of 1.05, shown as a histogram in fig. S12), they are comparable to the estimated size of measurement error (fig. S10). This suggests that most of the residual error observed in the antigenic map fit of the data are consistent with measurement error but that not all variation in the titers is captured.

We investigated whether any variant and serum group combinations exhibited consistently larger differences between fitted and measured titers (fig. S13). We find some variants showing differences in fitted versus measured titers in some serum groups, such as the B.1.617.2 variant having larger measured than fitted titers against the B.1.617.2 sera. As discussed in the main text, we speculate that when compared with other datasets (fig. S7), our measurements of B.1.617.2 serum antibodies against B.1.617.2 were biased toward higher values. Also as described in the main text, in cases such as the BA.4/BA.5 variant, a third dimension is required to better reflect the measured patterns

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of reactivity. However, some of these systematic residual errors in the map fit will also be reflective of the fact that not all patterns of cross-reactivity between the serum groups can be captured in the constraints of the Euclidean space of an antigenic map. In particular, some limitations may relate to the immunodominance differences identified in this paper, whereby one serum may see an antigenic difference between two variants and one serum may not. Antigenic cartography allows for this to some extent by permitting serum positions not to be coincident with homologous antigen positions. This means that a serum point can be positioned directly between two variants for which it sees no difference, thus being antigenically equidistant from both. However, this cannot be reconciled in cases where a serum has maximum reactivity to two variants and hence an expected antigenic distance of 0 between them. Overall though, we judge that the 2D map is a reasonable representation of the broader scale underlying antigenic relationships among variants, providing a useful complement to raw titer analysis, and in some cases may compensate for systematic titer biases like that of P.1 that are otherwise spurious.

Assessing the robustness of variant and serum positions to measurement error and presence or absence of variants and serum groups

We performed three experiments to determine the uncertainty of the positions of variants and sera in the antigenic map. First, we estimated how titration measurement error and variability in antigen reactivity would affect conclusions about antigenic map positions (fig. S14). We performed a "smooth" bootstrap with normally distributed noise added to the titer measurements and/or to antigen reactivity. Noise added to the titer measurement had a SD of 0.62, in keeping with the SD of repeat titrations shown in fig. S10, and random noise for antigen reactivity had a SD of 0.4, indicating that systematic biases in titers like that suspected for P.1 may also be present to some degree for other strains in the assay. We performed 500 repeats of adding random noise (using the Racmacs function bootstrapMap, method = "noisy") and summarized the variant positions by showing the area in which 68% (one SD) of the positional variation of a serum or antigen is captured. In general, we found that the antigen and serum positions are robust to both the random and systematic noise added, without affecting the main conclusions about the antigenic relationships between the variants included.

Next, we assessed whether the positions of the variants are robust to the exclusion of each variant in turn. We find that positions of variants are robust to the exclusion of variants (fig. S15), with the small movement observed with the exclusion of D614G and B.1.351. We then investigated whether positions of variants are robust to the exclusion of each serum group in turn (fig. S16). Overall, the map was robust to the exclusion of single serum groups, with the greatest movement observed when excluding the 2× mRNA-1273.351 vaccine serum group and the B.1.617.2 and BA.1 convalescent serum groups. Relatedly, since we incorporated the information from the different post–mRNA-1273 vaccine serum groups into the map, we also explored how the antigenic map compared if constructed from postinfection sera alone (fig. S17). For some variants that had not been titrated against post-D614G infection sera (and were thus less well coordinated without including post-D614G vaccination sera titrations), we did see larger discrepencies in antigenic position in the postinfection sera–only map, in particular for the BA.4/BA.5 variant. (fig. S17A). For other variants, antigenic relationships were consistently well described in the presence or absence of postvaccination sera (fig. S17B). We also included the outlier sera identified in fig. S2 and found little difference in map topology (fig. S19).

We also evaluated the robustness of map positions on particular titrations by performing 500 "Bayesian bootstrap" repeats, where in each repeat, titers were reweighted according to a dirichlet distribution. This method is statistically similar to the more common method of bootstrapping using random sampling with replacement, but it ensures that titers are never fully removed from the dataset, which can happen when sampling with replacement and lead to underconstrained antigenic map solutions in some of the bootstrap repeats. We performed this reweighting on both variants and sera combined, as well as antigens and sera separately, using the Racmacs function bootstrapMap, method = "bayesian" (fig. S20). Map positions were largely robust to this reweighted sampling of titers.

We also show that the error is approximately equally distributed between all variants and sera (fig. S21A). Positions of variants and sera are also robust when allowing the variant and sera to move in the map while not increasing the overall map root mean square error (RMSE) between map and table distance by more than one unit (fig. S21B).

Assessing predictive power of the antigenic map (cross-validation)

To investigate whether the antigenic map is able to predict missing titers, we performed 500 cross-validation replicates, where in each replicate we removed 10% of all titrations, and predicted the missing titers from the resulting map. On the $\log_2$ scale, where difference of 1 represents a predicted titer that is 2-fold higher than the measured titer and −1 represents a titer that is twofold lower, the mean difference between predicted and measured titer was 0.53 with a SD of 1.61 (fig. S22). We also considered the difference between predicted and measured titers split by variant and serum group (fig. S23). As for figs. S12 and S13, these results imply that not all patterns of cross-reactivity between the serum groups can be captured in the constraints of the 2D Euclidean space of the map but that overall, the 2D relationships are largely predictive of titers among sera and variants.

Construction of the antibody landscapes

The construction of the antibody landscapes differs from the methodology used in previous publications (*34*, *71*) , since each serum represents a likely first infection and was used to construct the map itself with explicit assumptions about how reactivity should be distributed. Given a serum with coordinates (x_m, y_m), let t_m be its maximum $\log_2$ titer/10. Then (x_m, y_m, t_m) represents the highest point of the antibody landscape for this serum. Given any other point $p = (x, y)$ on the 2D base space, the third coordinate, t, of the surface at this point is given by

$$t = t_m - d(p, p_m)$$

where $d(p, p_m)$ is the Euclidean distance between p and p_m. In particular, the serum point and maximum titer for a given serum describe how reactivity would be expected to vary in a cone-like fashion across antigenic space, with its apex at the serum point at a height equal to the maximum measured $\log_2$ titer/10 and diminishing at a rate of a twofold reduction in titer with each antigenic unit on the map. The serum group averages shown in Fig. 4 therefore simply represent an average of all these individual slope = 1 cone-like responses for the relevant sera, displaying the GMT of all the individual titer predictions for the group.

In this case, since samples were collected after a presumed single exposure, individual landscapes simply visualize the slope 1 cone-like model of reactivity that is fit when constructing the antigenic map, rather than refitting the same data with a nonparametric modeling approach that can also be applied.

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ACKNOWLEDGMENTS

We thank A. Eaton, E. Domin, S. Hiles, J. Tong, L. Liu, K. Lund, J. Chen, K. Bradley, and W. Feng, Duke University, for immunological analysis. We thank our collaborators on specimen collection and coordination: S. Kaweski, C. Marquez, BC Centre for Disease Control; the Community Surveillance for Infection Team, Epidemiology Task Force, COVID-19 Response (M. Rolfes and A. Dixon), and the Genomics and Discovery Team, Respiratory

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Virus Branch DVD, NCIRD from the US Centers for Disease Control and Prevention; the Coronavirus Household Evaluation and Respiratory Testing (C-HEaRT) Cohort Columbia University team (P. Thind, M. Castro, F. Sosa, and L. Wang) and Abt Associates team (J. Duque, Z. Jeddy, K. Altunkaynak, M. Duckworth, and U. Kattel); the CoVPN (COVID-19 Prevention Network) protocol 5001 team; the Natural History of SARS-CoV-2 and Influenza in Peru Study Team; C. Barkauskas, S. Wolery, and I. Naqvi, Duke University Medical Center; the Personalized Virology Initiative at the Icahn School of Medicine at Mount Sinai (D. Altman, M. C. Bermúdez-González, M. Saksena, K. Srivastava, C. Gleason, G. Kleiner, and A. Wajnberg) and the Mount Sinai Pathogen Surveillance Program (A. S. Gonzalez-Reiche, E. M. Sordillo, and A. Paniz-Mondolfi); and J. Beigel and P. C. Roberts from the Division of Microbiology and Infectious Diseases, NIAID, NIH, Bethesda, MD, USA. We are deeply grateful for the generous support of all study participants. The findings and conclusions in this report are those of the author(s) and do not necessarily represent the official position of the [US] Centers for Disease Control and Prevention (CDC). **Funding:** S.H.W., S.T., E.B.L., A.N., R.A.M.F., T.C.J., and D.J.S.: Centers of Excellence for Influenza Research and Response (CEIRR, contract no. 75N93021C00014) as part of the SARS-CoV-2 Assessment of Viral Evolution (SAVE) Program. X.S., D.C.M., C.M., H.T., and L.D.: SAVE Program (contract no. 75N93019C00050 Opt 18A and 18B). B.M., T.C.J., C.D., and V.M.C.: Funded by the German Federal Ministry of Education and Research through project DZIF (8040701710 and 8064701703) and the German Federal Ministry of Health through project SeroVarCoV. V.M.C. and L.M.J.: Funded by the German Federal Ministry of Education and Research through project VARIpath (01KI2021). H.v.B., V.S., and F.K.: NIAID CEIRR contract no. 75N93021C00014, SAVE option 12A, and the NIAID Collaborative Influenza Vaccine Innovation Centers (CIVIC) contract no. 75N93019C00051. E.B.L. is funded by a Gates Cambridge Scholarship. A.N. was supported by the Gates Cambridge Trust. R.P.: Moderna. Also supported by the Office of the Assistant Secretary for Preparedness and Response, Biomedical Advanced Research and Development Authority (contract no. 75A50120C00034) and by NIAID. Y.K.: Research funding support to UW-Madison and University of Tokyo. Center for Research on Influenza Pathogenesis and Transmission (CRIPT) (75N93021C00014) and the Japan Program for Infectious Diseases Research and Infrastructure (JP21wm0125002) from the Japan Agency for Medical Research and Development (AMED). M.S.S.: Funded by the CDC (contract no. 75D30120C08150 with Abt Associates). **Author contributions:** Conceptualization: S.H.W., B.M., X.S., D.C.M., and D.J.S. Methodology: S.H.W. and D.J.S. Software: S.H.W. and T.C.J. Validation: S.H.W., B.M., X.S., S.T., E.B.L., A.N., D.C.M., and D.J.S. Formal analysis: S.H.W., B.M., X.S., S.T., E.B.L., and A.N. Investigation: S.H.W., B.M., X.S., S.T., E.B.L., A.N., D.C.M., and D.J.S. Resources: X.S., M.A.C., J.N.C., M.B.D., T.N.D., C.D., R.A.M.F., P.J.G., V.M.C., L.J., P.J.H., A.J., Y.K., F.K., R.P., V.S., M.S., F.S.D., V.V., H.v.B., R.W., D.C.M., D.J.S., L.M.J., V.M.C., and C.D. Data curation: S.H.W., B.M., and X.S. Writing - Original Draft: S.H.W., B.M., and D.J.S. Writing - Review and Editing: All. Visualization: S.H.W., B.M., S.T., A.N., and E.B.L. Supervision: D.C.M. and D.J.S. **Competing interests:** V.M.C. is named on patents regarding SARS-CoV-2 serological testing and monoclonal antibodies. He is also a part-time employee at Labor Berlin - Charité Vivantes GmbH, a diagnostic laboratory and subsidiary of Charité - Universitätsmedizin Berlin and the Vivantes – Netzwerk für Gesundheit GmbH. F.K. has been consulting for Curevac, Seqirus, and Merck, and is currently consulting for Pfizer, Third Rock Ventures, Avimex, and GSK. He is named on several patents regarding influenza virus and SARS-CoV-2 virus vaccines, influenza virus therapeutics, and SARS-CoV-2 serological tests. Some of these technologies have been licensed to commercial entities, and F.K. is receiving royalties from these entities. F.K. is also an advisory board member of Castlevax, a spin-off company formed by the Icahn School of Medicine at Mount Sinai to develop SARS-CoV-2 vaccines. The Krammer laboratory has received funding for research projects from Pfizer, GSK, and Dynavax, and three of F.K.'s mentees have recently joined Moderna. Y.K. has research funding from FUJIFILM Toyama Chemical Co. (various viruses), Shionogi & Co. (influenza viruses), Daiichi Sankyo Pharmaceutical, Otsuka Pharmaceutical, KM Biologics, Kyoritsu Seiyaku, Fuji Rebio, and Tauns Laboratories. He is a FluGen cofounder and a stockholder of FluGen. **Data and materials availability:** Serum samples from individuals infected with SARS-CoV-2 are available from A. Jassem under a material transfer agreement with the University of British Columbia and the Provincial Health Services Authority. Code and data to reproduce the results and figures in this paper are available from Zenodo (*56*). **License information:** Copyright © 2023 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse. This research was funded in whole or in part by US govt contract # 75N93021C00014 and the Gates Cambridge Trust, a cOAlition S organizations. The author will make the Author Accepted Manuscript (AAM) version available under a CC BY public copyright license.

SUPPLEMENTARY MATERIALS

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Submitted 5 June 2023; accepted 23 August 2023
10.1126/science.adj0070

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Mapping SARS-CoV-2 antigenic relationships and serological responses

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Science **382** (6666), eadj0070. DOI: 10.1126/science.adj0070

Editor's summary

The SARS-CoV-2 pandemic was marked by waves of new strains of virus differing in virulence and immune reactivity. The advent of each new variant of concern brought more human casualties and waves of onerous quarantine measures. To map the evolutionary trajectory of the variants, Wilks *et al*. obtained sera from people who had been vaccinated or infected with a range of variants of concern and applied antigenic cartography to visualize structural changes in the virus. The authors observed changes in immunodominance and immune escape depending on the variant that had infected the patient or after vaccination. Such analysis has implications for variant risk assessment and for selecting the next candidate vaccine strains that will confer the highest protection. —Caroline Ash

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